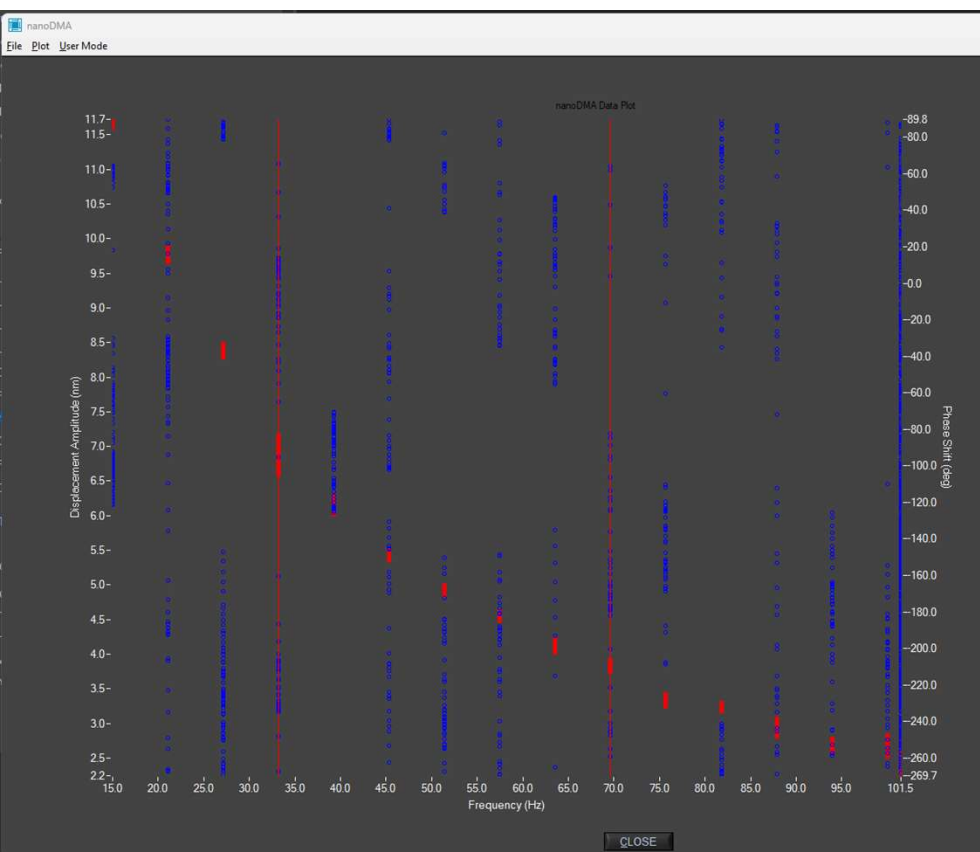




Screenshot1

Load Function Type: Frequency Sweep

Feedback Type: Displacement Control

Parameters

Peak Displacement (nm): 250.000

Displacement Amplitude (nm): 10.000

Start Frequency (Hz): 100.000

End Frequency (Hz): 15.000

Number Of Steps: 15

Frequency Scaling: Linear

Generate Load Function

Gain Levels: Level 5

Lower Frequency Region Below(Hz): 10.0

Lock

Default

Upto Frequency: 0.99 Hz, 3 Hz, 10 Hz, 30 Hz, 100 Hz, 1000 Hz

Cutoff Ratio Values: 1.0E+002, 7.0E+000, 5.0E-001, 3.0E-001, 7.0E-002, 7.5E-002

Parameter: Gain Settings

Upper Frequency: Lower Frequency

Quasi Static

Proportional: 1.0000, 1.5000

Integral: 1.0000, 0.2000

Derivative: 0.0000

nanoDMA

Proportional DMA: 0.0000, 1.5000

Integral DMA: 5.0000, 0.2000

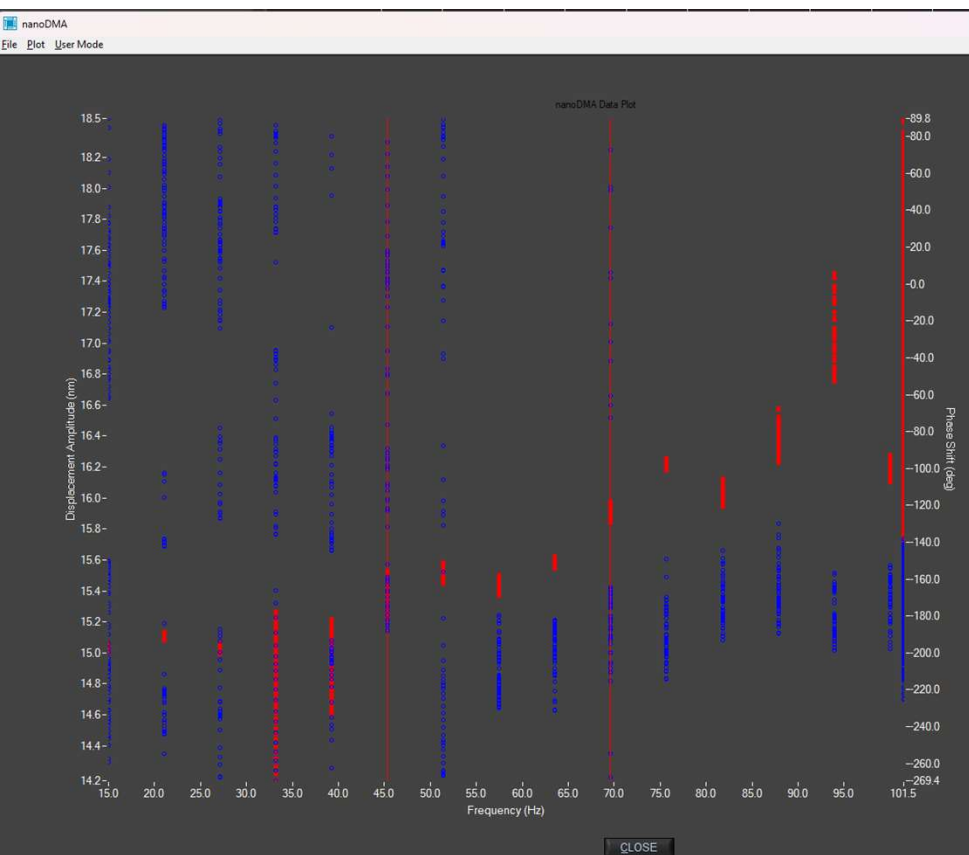
Derivative DMA: 0.0000

Parameter

Stiffness Correction: 0.0000

Number Points Average: 0.0000

Slope Adjustment: 1.0000



Screenshot2

Load Function Type: **Frequency Sweep**

Feedback Type: **Displacement Control**

Parameters

- Peak Displacement (nm): 250.000
- Displacement Amplitude (nm): 10.000
- Start Frequency (Hz): 100.000
- End Frequency (Hz): 15.000
- Number Of Steps: 15
- Frequency Scaling: **Linear**

Generate Load Function

Gain Levels: **Level 5** [Lock] [Default]

Lower Frequency Region Below(Hz): 10.0

Upto Frequency: 0.99 Hz 3 Hz 10 Hz 30 Hz 100 Hz 1000 Hz

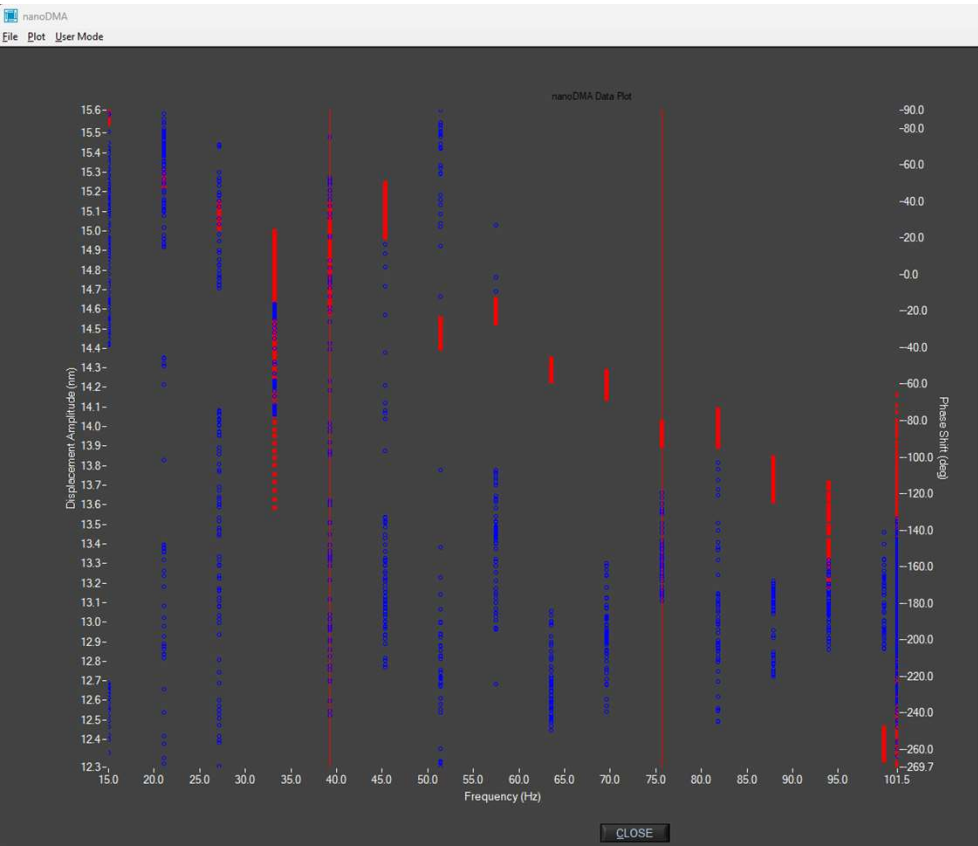
Cutoff Ratio Values: **1.0E+002** **7.0E+000** **5.0E-001** **3.0E-001** **7.0E-002** **7.5E-002**

Parameter Gain Settings

	Upper Frequency	Lower Frequency
Quasi Static		
Proportional	0.0000	1.5000
Integral	10.0000	0.2000
Derivative	0.0000	
nanoDMA		
Proportional DMA	0.0000	1.5000
Integral DMA	5.0000	0.2000
Derivative DMA	0.0000	

Parameter

- Stiffness Correction: 0.0000
- Number Points Average: 0.0000
- Slope Adjustment: 1.0000



Screenshot3

Load Function Type: **Frequency Sweep**

Feedback Type: **Displacement Control**

Parameters

Peak Displacement (nm): **250.000**

Displacement Amplitude (nm): **10.000**

Start Frequency (Hz): **100.000**

End Frequency (Hz): **15.000**

Number Of Steps: **15**

Frequency Scaling: **Linear**

Generate Load Function

Gain Levels: **Level 5** **Lock** **Default**

Lower Frequency Region Below(Hz): **10.0**

Upto Frequency: **0.99 Hz** **3 Hz** **10 Hz** **30 Hz** **100 Hz** **1000 Hz**

Cutoff Ratio Values: **1.0E+002** **7.0E+000** **5.0E-001** **3.0E-001** **7.0E-002** **7.5E-002**

Parameter: **Gain Settings**

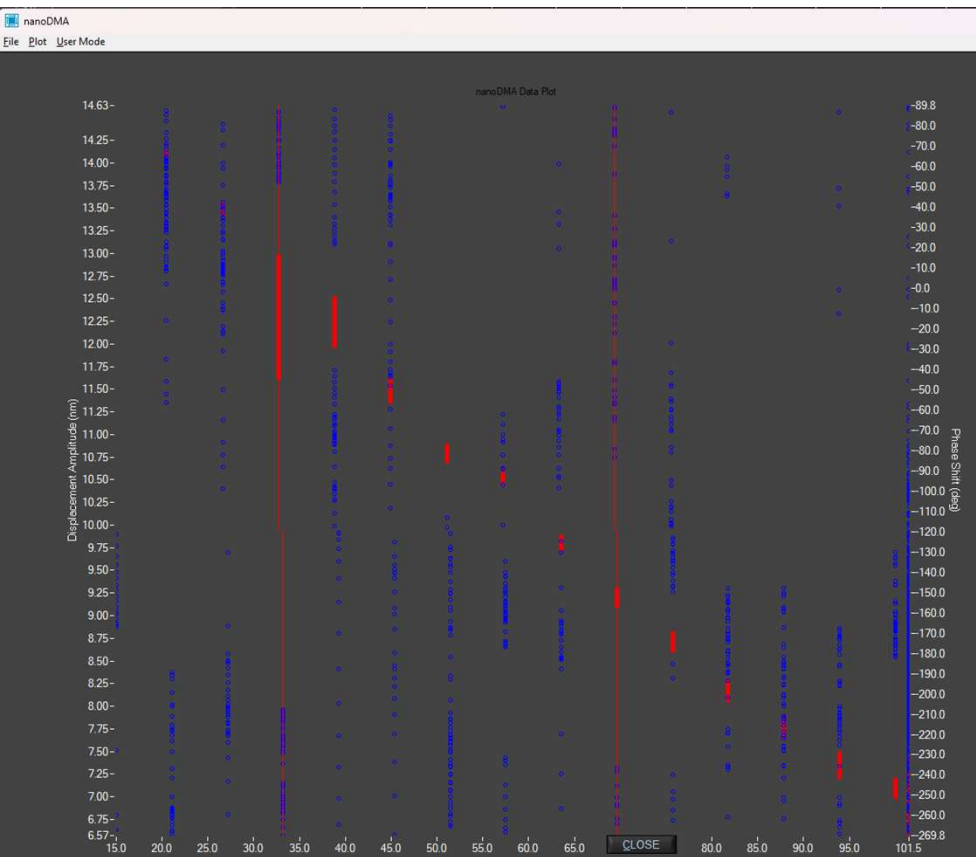
	Upper Frequency	Lower Frequency
Quasi Static		
Proportional	0.0000	1.5000
Integral	5.0000	0.2000
Derivative	0.0000	
nanoDMA		
Proportional DMA	0.0000	1.5000
Integral DMA	5.0000	0.2000
Derivative DMA	0.0000	

Parameter

Stiffness Correction: **0.0000**

Number Points Average: **0.0000**

Slope Adjustment: **1.0000**



Screenshot4

Load Function Type **Frequency Sweep**

Feedback Type **Displacement Control**

Parameters

Peak Displacement (nm) **250.000**

Displacement Amplitude (nm) **10.000**

Start Frequency (Hz) **100.000**

End Frequency (Hz) **15.000**

Number Of Steps **15**

Frequency Scaling **Linear**

Generate Load Function

Gain Levels **Level 5** **Lock** **Default**

Lower Frequency Region Below(Hz) **10.0**

Upto Frequency **0.99 Hz** **3 Hz** **10 Hz** **30 Hz** **100 Hz** **1000 Hz**

Cutoff Ratio Values **1.0E+002** **7.0E+000** **5.0E-001** **3.0E-001** **7.0E-002** **7.5E-002**

Parameter **Gain Settings**

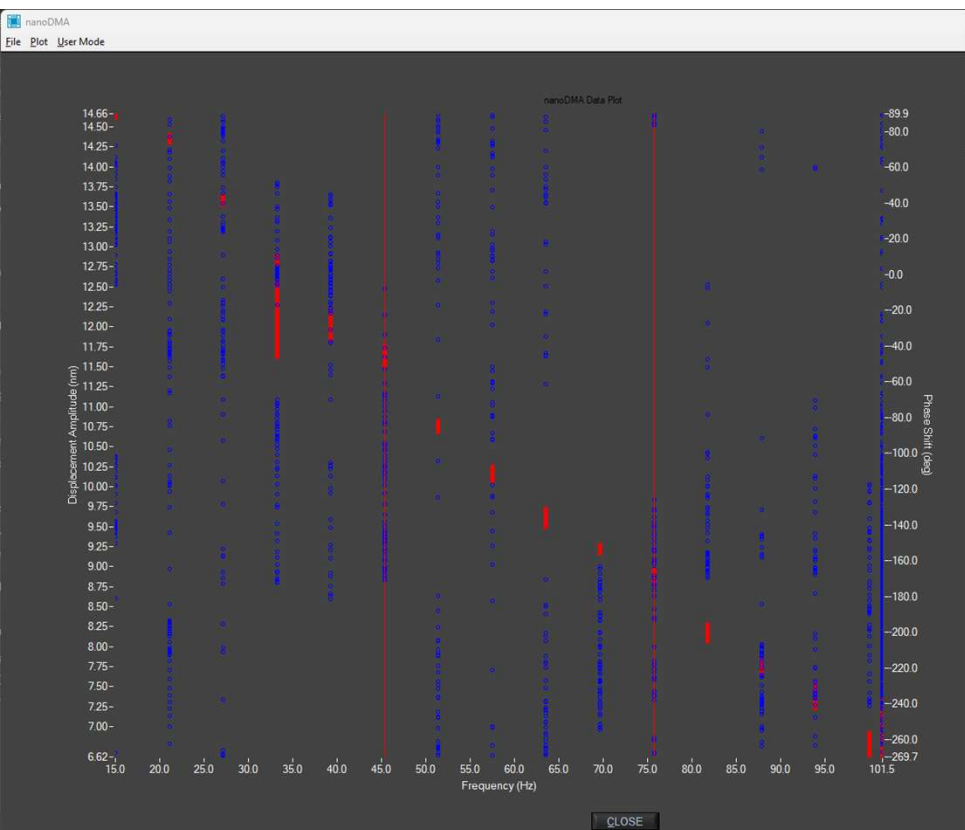
	Upper Frequency	Lower Frequency
Quasi Static		
Proportional	0.0000	1.5000
Integral	2.5000	0.2000
Derivative	0.0000	
nanoDMA		
Proportional DMA	0.0000	1.5000
Integral DMA	5.0000	0.2000
Derivative DMA	0.0000	

Parameter

Stiffness Correction **0.0000**

Number Points Average **0.0000**

Slope Adjustment **1.0000**



Screenshot5

Load Function Type **Frequency Sweep**

Feedback Type **Displacement Control**

Parameters

Peak Displacement (nm) **250.000**

Displacement Amplitude (nm) **10.000**

Start Frequency (Hz) **100.000**

End Frequency (Hz) **15.000**

Number Of Steps **15**

Frequency Scaling **Linear**

Generate Load Function

Gain Levels **Level 5** **Lock** **Default**

Lower Frequency Region Below(Hz) **10.0**

Upto Frequency **0.99 Hz** **3 Hz** **10 Hz** **30 Hz** **100 Hz** **1000 Hz**

Cutoff Ratio Values **1.0E+002** **7.0E+000** **5.0E-001** **3.0E-001** **7.0E-002** **7.5E-002**

Parameter

Gain Settings

Upper Frequency Lower Frequency

Quasi Static

Proportional **0.0000** **1.5000**

Integral **2.5000** **0.2000**

Derivative **0.0000**

nanodMA

Proportional DMA **0.0000** **1.5000**

Integral DMA **10.0000** **0.2000**

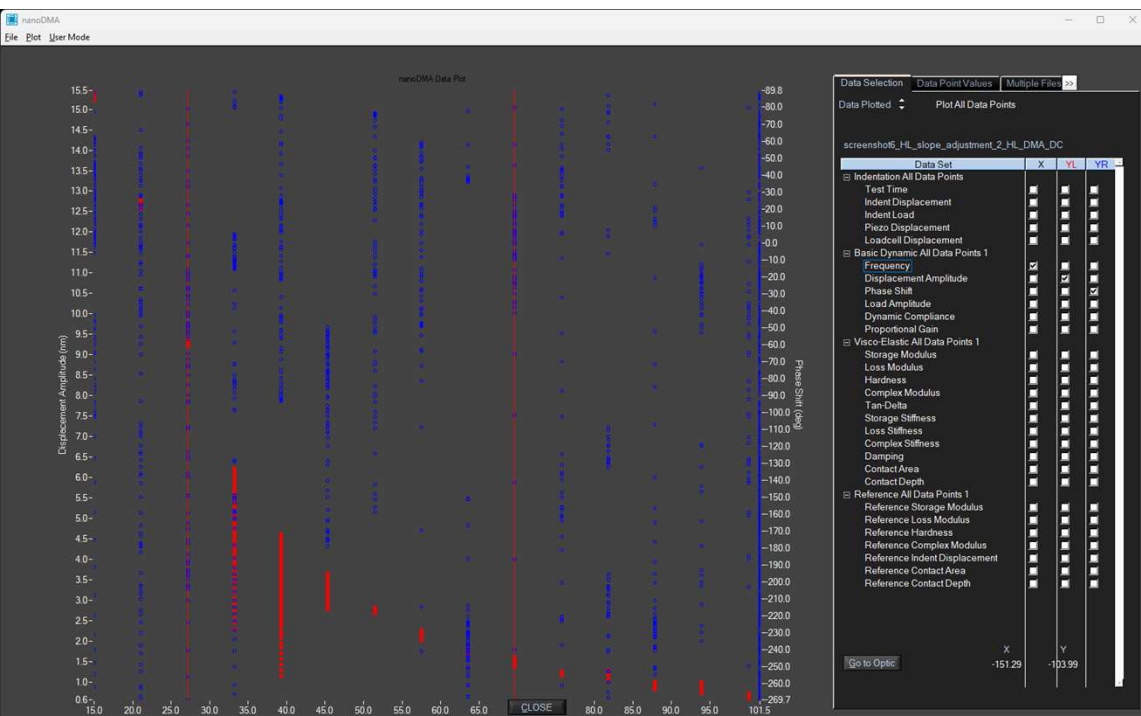
Derivative DMA **0.0000**

Parameter

Stiffness Correction **0.0000**

Number Points Average **0.0000**

Slope Adjustment **1.0000**



Data Selection Data Point Values Multiple Files

Data Plotted Plot All Data Points

screenShot6_HL_slope_adjustment_2_HL_DMA_DC

Data Set	X	Y	YR
Indentation All Data Points			
Test Time			
Indent Displacement			
Indent Load			
Piezo Displacement			
Loadcell Displacement			
Basic Dynamic All Data Points 1			
Frequency			
Displacement Amplitude			
Phase Shift			
Load Amplitude			
Dynamic Compliance			
Proportional Gain			
Visco-Elastic All Data Points 1			
Storage Modulus			
Loss Modulus			
Hardness			
Complex Modulus			
Tan-Delta			
Storage Stiffness			
Loss Stiffness			
Complex Stiffness			
Damping			
Contact Area			
Contact Depth			
Reference All Data Points 1			
Reference Storage Modulus			
Reference Loss Modulus			
Reference Hardness			
Reference Complex Modulus			
Reference Indent Displacement			
Reference Contact Area			
Reference Contact Depth			

Go to Optics

X -151.29 Y -101.99

Screenshot6

Load Function Type Frequency Sweep

Feedback Type Displacement Control

Gain Levels Level 5 Lock

Lower Frequency Region Below(Hz) 10.0 Default

Upto Frequency 0.99 Hz 3 Hz 10 Hz 30 Hz 100 Hz 1000 Hz

Cutoff Ratio Values 1.0E-002 7.0E-000 5.0E-001 3.0E-001 7.0E-002 7.5E-002

Parameters

Peak Displacement (nm) 250.000

Displacement Amplitude (nm) 1.000

Start Frequency (Hz) 100.000

End Frequency (Hz) 15.000

Number Of Steps 15

Frequency Scaling Linear

Generate Load Function

Parameter Gain Settings

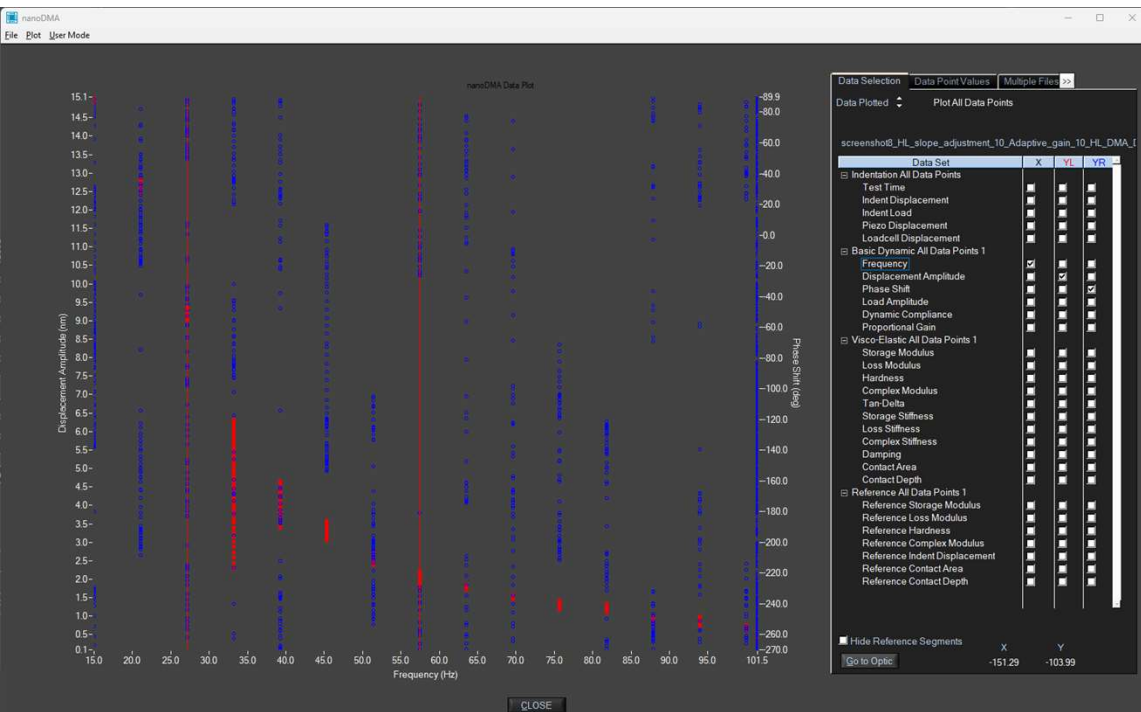
	Upper Frequency	Lower Frequency
Quasi Static		
Proportional	0.0000	1.5000
Integral	1.0000	0.2000
Derivative	0.0000	
nanoDMA		
Proportional DMA	0.0000	1.5000
Integral DMA	5.0000	0.2000
Derivative DMA	0.0000	

Parameter

Stiffness Correction 0.0000

Number Points Average 0.0000

Slope Adjustment 2.0000



Screenshot7

Load Function Type: Frequency Sweep

Feedback Type: Displacement Control

Parameters

Peak Displacement (nm): 250.000

Displacement Amplitude (nm): 1.000

Start Frequency (Hz): 100.000

End Frequency (Hz): 15.000

Number Of Steps: 15

Frequency Scaling: Linear

Generate Load Function

Gain Levels: Level 5

Lower Frequency Region Below(Hz): 10.0

Lock

Default

Upto Frequency: 0.99 Hz 3 Hz 10 Hz 30 Hz 100 Hz 1000 Hz

Cutoff Ratio Values: 1.0E+002 7.0E+000 5.0E-001 8.0E-001 7.0E-002 7.5E-002

Parameter Gain Settings

	Upper Frequency	Lower Frequency
Quasi Static		
Proportional	0.0000	1.5000
Integral	1.0000	0.2000
Derivative	0.0000	
nanoDMA		
Proportional DMA	0.0000	1.5000
Integral DMA	5.0000	0.2000
Derivative DMA	0.0000	

Parameter

Stiffness Correction: 0.0000

Number Points Average: 0.0000

Slope Adjustment: 2.0000



Screenshot9

Load Function Type: **Frequency Sweep**

Feedback Type: **Displacement Control**

Parameters

Peak Displacement (nm): **250.000**

Displacement Amplitude (nm): **1.000**

Start Frequency (Hz): **100.000**

End Frequency (Hz): **15.000**

Number Of Steps: **15**

Frequency Scaling: **Linear**

Generate Load Function

Gain Levels: **Level 5** **Lock** **Default**

Lower Frequency Region Below(Hz): **10.0**

Upto Frequency: **0.99 Hz** **3 Hz** **10 Hz** **30 Hz** **100 Hz** **1000 Hz**

Cutoff Ratio Values: **1.0E+002** **7.0E+000** **5.0E-001** **3.0E-001** **7.0E-002** **7.5E-002**

Parameter: **Gain Settings**

	Upper Frequency	Lower Frequency
Quasi Static		
Proportional	0.0000	1.5000
Integral	0.1000	0.2000
Derivative	0.0000	
nanoDMA		
Proportional DMA	0.0000	1.5000
Integral DMA	0.1000	0.2000
Derivative DMA	0.0000	

Parameter

Stiffness Correction: **0.0000**

Number Points Average: **0.0000**

Slope Adjustment: **2.0000**